NestHR backend: new server setup

Ubuntu server, from a blank machine to a live HTTPS API. Run every command in order. Replace the placeholders in <angle brackets>.

Item	Value
New server	192.168.1.121 (user: nesthr, needs sudo)
Old server	192.168.1.131 (user: laabhbothra), app folder `~/server/kalahanu-server/hrms-pixelate`
Git repo	https://github.com/labyeye/hrms-pixelate.git (branch main)
Domain	hrms-backend.pixelatenest.com
App port	4000 (from PORT in backend/.env)
Node / Python	Node 22. Old server has Python 3.12; the new one has Python 3.14 (face service notes in Step 8)

Read first. The `.env` points to the live database (MONGO_URI) and the live WhatsApp number. If the old and new servers both run at the same time, the hourly attendance cron and WhatsApp messages will fire twice. So test the new server first, then switch DNS, then stop the old app (Step 14).

Copy-paste tip. A long command can get cut in the middle when the terminal wraps the line (it breaks paths like `.env` into `.e` + `nv`). Every command below is kept short and is safe to paste as-is. If a command still wraps, use the shorter variable form shown. Never press Enter in the middle of a command.

Step 1. Log in and update the system

```
ssh nesthr@192.168.1.121
sudo apt update && sudo apt -y upgrade
sudo timedatectl set-timezone Asia/Kolkata
```

If the SSH client says "REMOTE HOST IDENTIFICATION HAS CHANGED" on your PC, the IP was used by another machine before. Only if you are sure this is the new server, run: `ssh-keygen -R 192.168.1.121` and connect again.

Step 2. Install git and basic tools

```
sudo apt -y install git curl wget build-essential g++ cmake ufw rsync unzip
git --version
g++ --version | head -1
```

Step 3. Install Node 22 and pm2

```
curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
sudo apt -y install nodejs
node -v      # should print v22.x
sudo npm install -g pm2
pm2 -v
```

Step 4. Clone the repository

```
mkdir -p ~/server && cd ~/server
git clone https://github.com/labyeye/hrms-pixelate.git
cd hrms-pixelate
git log --oneline | head -3
```

If the repo is private, git asks for a username and password. Use your GitHub username and a Personal Access Token (not your GitHub password).

Step 5. Install backend packages

```
cd ~/server/hrms-pixelate/backend
npm ci --omit=dev
mkdir -p uploads backups
```

Only the backend runs on this server. The web app (frontend) is hosted separately, so you do not build it here.

Step 6. Copy the .env from the old server

Run this **on the new server**. The file travels directly between the two servers, so the secrets never pass through your PC. The password asked is the **old server's** (laabhbothra) password.

```
cd ~/server/hrms-pixelate/backend
OLD=laabhbothra@192.168.1.131:/home/laabhbothra/server/kalahanu-server/hrms-pixelate/backend
scp $OLD/.env .env
chmod 600 .env
sed 's/=.*/=***/' .env
```

Use the **full path** `/home/laabhbothra/...`. Do not write `~/server/...` for the old-server part: your shell replaces `~` with the new server's home (/home/nesthr) before the command is sent, and the file is then "not found".

You should see these 13 keys: PORT, MONGO_URI, JWT_SECRET, NODE_ENV, RAZORPAY_KEY_ID, RAZORPAY_KEY_SECRET, META_WA_TOKEN, META_WA_PHONE_ID, META_WA_VERIFY_TOKEN,

CRM_API_SECRET, CRM_SUPPORT_WEBHOOK_URL, FACE_SERVICE_URL, FACE_SERVICE_API_KEY.

Add the keys the new setup needs

```
cd ~/server/hrms-pixelate/backend
echo 'ALLOWED_ORIGINS=https://hrms.pixelatenest.com' >> .env
tail -3 .env | sed 's/=.*/=***/'
```

Required. pm2 runs the app with NODE_ENV=production, and in production the backend refuses to start without `ALLOWED_ORIGINS` (the log shows "[FATAL] ALLOWED_ORIGINS must be set in production"). Put the web app addresses here, separated by commas with no spaces. Mobile apps and the CRM dashboard do not need to be listed.
Check that the new line is on its own line (the old file may not end with a newline). If it got glued to the last line, fix it with `nano .env`.

Also check (nano .env): `PORT=4000`; `FACE_SERVICE_URL=http://127.0.0.1:8091`; if MONGO_URI is a cloud database (Atlas), allow the new server's public IP in its network access list.
Recommended: add `META_WA_APP_SECRET=<Meta App Secret>` (Meta developer dashboard). Without it the app logs "incoming webhooks are NOT signature-verified".

Step 7. Quick test before pm2

```
cd ~/server/hrms-pixelate/backend
node server.js > /tmp/hrms.log 2>&1 &
sleep 8
curl -s -m 5 http://localhost:4000/api/health; echo
tail -15 /tmp/hrms.log
kill %1
```

Expected: `{"status":"ok","service":"HRMS API"}` and the log lines "HRMS server running on port 4000" and "MongoDB connected". If it fails with a database error, fix MONGO_URI / the IP allow-list before moving on.

This test connects to the live database, so the hourly attendance auto-mark job runs once. That is safe (it does not create duplicates). The log can show the database host name; do not paste it publicly.

Step 8. Face service (Python)

Needed only for face-recognition attendance. If you do not use it, skip this step and, in Step 9, start only the API with `pm2 start ecosystem.config.js --only hrms-backend`.

```
sudo apt -y install python3-venv python3-dev cmake g++ libopenblas-dev liblapack-dev
cd ~/server/hrms-pixelate/face-service
python3 -m venv venv
venv/bin/pip install --upgrade pip
venv/bin/pip install dlib-bin==20.0.1
venv/bin/pip install --no-deps -r requirements.txt
venv/bin/python -c "import dlib, face_recognition; print('face ok')"
```

Why this way: requirements.txt lists `dlib-bin` and `face-recognition`, and face-recognition would otherwise try to build the real `dlib` from source (error: "No CMAKE_CXX_COMPILER could be found"). Installing dlib-bin first and then using `--no-deps` avoids that, because every package is already pinned in the file.
If `dlib-bin` has no wheel for the new Python (3.14), run `venv/bin/pip install dlib` instead (builds from source, 10 to 20 minutes), then the `--no-deps` line again. If that also fails, create the venv with Python 3.12, as on the old server.

If the old server keeps face data on disk, copy it too:

```
OLD=laabhbothra@192.168.1.131:/home/laabhbothra/server/kalahanu-server/hrms-pixelate
rsync -avz --exclude venv --exclude __pycache__ $OLD/face-service/ ~/server/hrms-pixelate/face-service/
```

Step 9. Start with pm2

`ecosystem.config.js` starts the API in **cluster mode with 4 workers** (that is why you see 4 lines named hrms-backend: they are 4 copies of the same app sharing port 4000) plus the face service. The face service needs the same key as `FACE_SERVICE_API_KEY` in .env.

```
cd ~/server/hrms-pixelate
grep FACE_SERVICE_API_KEY backend/.env          # copy this value
nano ecosystem.config.js                         # replace CHANGE_ME with that value, save: Ctrl+O, Enter, Ctrl+X
pm2 start ecosystem.config.js
sleep 8
pm2 status
curl -s -m 5 http://localhost:4000/api/health; echo
```

All 4 `hrms-backend` workers should be `online` with restart count 0. Make pm2 start after every reboot:

```
pm2 save
pm2 startup systemd -u nesthr --hp /home/nesthr
# pm2 prints one "sudo env PATH=... pm2 startup ..." line. Copy that exact line, run it, then:
pm2 save
```

Workers. 4 workers use about 4x the memory (roughly 100 to 150 MB each). Check `nproc` and `free -m`. For a small server set `instances: 2`, or for a single process `instances: 1, exec_mode: "fork"` in ecosystem.config.js. The hourly attendance cron runs only on worker 0.

If workers show "errored" and keep restarting: `pm2 logs hrms-backend --lines 40 --nostream --err`. Common causes: "ALLOWED_ORIGINS must be set in production" (Step 6), a missing module (`cd backend && npm ci --omit=dev`), or database access (Step 6 note). After fixing .env run `pm2 restart hrms-backend --update-env`. To reset the restart counter: `pm2 delete hrms-backend`, then `pm2 start ecosystem.config.js --only hrms-backend`.

Step 10. Copy uploaded files from the old server

```
cd ~/server/hrms-pixelate/backend
SRC=laabhbothra@192.168.1.131:/home/laabhbothra/server/kalahanu-server/hrms-pixelate/backend
rsync -avz $SRC/uploads/ uploads/
rsync -avz $SRC/backups/ backups/
du -sh uploads backups
```

Run the `SRC=...` line in one piece, then `echo $SRC` to confirm the full address shows. The password asked is the old server's. Encrypted document-vault files live in uploads, so do not skip this if employees uploaded documents. If `backups` is very large (several GB) you can skip it; those are only old backup copies.

Step 11. Install and configure nginx

```
sudo apt -y install nginx
sudo nano /etc/nginx/sites-available/nesthr
```

Paste this (HTTP only; certbot adds HTTPS in Step 13):

```
server {
    listen 80;
    server_name hrms-backend.pixelatenest.com;

    # ESSL / ZKTeco biometric device push
    location /iclock/ {
        proxy_pass http://127.0.0.1:4000;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 60s;
    }

    location /api/ {
        client_max_body_size 10m;
        proxy_read_timeout 60s;

        if ($request_method = OPTIONS) {
            add_header 'Access-Control-Allow-Origin' $http_origin always;
            add_header 'Access-Control-Allow-Methods' 'GET, POST, PUT, DELETE, PATCH, OPTIONS'
always;
            add_header 'Access-Control-Allow-Headers' 'Authorization, Content-Type, Accept' alw
ays;
            add_header 'Access-Control-Allow-Credentials' 'true' always;
            add_header 'Access-Control-Max-Age' 86400 always;
            return 204;
        }

        proxy_pass http://127.0.0.1:4000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
        proxy_cache_bypass $http_upgrade;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

Enable it and reload:

```
sudo ln -s /etc/nginx/sites-available/nesthr /etc/nginx/sites-enabled/nesthr
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo systemctl reload nginx
sudo systemctl enable nginx
curl -s -H "Host: hrms-backend.pixelatenest.com" http://127.0.0.1/api/health
```

Step 12. Firewall

```
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw enable
sudo ufw status
```

> Do not open port 4000 or 8091 to the outside; only nginx should reach them.

Step 13. HTTPS with certbot

Before this step the DNS record `hrms-backend.pixelatenest.com` must point to this server's public IP, and ports 80 and 443 must be reachable from the internet (router / cloud firewall).

```
sudo apt -y install snapd
sudo snap install core && sudo snap refresh core
sudo snap install --classic certbot
sudo ln -sf /snap/bin/certbot /usr/bin/certbot
sudo certbot --nginx -d hrms-backend.pixelatenest.com
sudo certbot renew --dry-run
```

Certbot edits the nginx file to add the 443 block and the HTTP-to-HTTPS redirect. Auto-renew is installed by certbot itself.

Final test: `curl -s https://hrms-backend.pixelatenest.com/api/health`

Step 14. Switch over from the old server

1. Point the DNS record to the new server (if not done already). Wait until `nslookup hrms-backend.pixelatenest.com` shows the new IP.
2. Check the web app and mobile app work (login, attendance, payroll page).
3. Check a biometric device still pushes punches (pm2 logs show `/iclock` requests).
4. Only then stop the app on the OLD server, so the cron and WhatsApp do not run twice:

   ```
   ssh laabhbothra@192.168.1.131
   pm2 stop hrms-pixelate face-service
   pm2 save
   ```

5. Keep the old copy for a few days as a backup, then delete it.

Daily commands

Task	Command
See status	`pm2 status`
Live logs	`pm2 logs hrms-backend`

Update the app	`cd ~/server/hrms-pixelate && git pull && cd backend && npm ci --omit=dev && pm2 reload hrms-backend`
Restart	`pm2 restart hrms-backend`
nginx test / reload	`sudo nginx -t && sudo systemctl reload nginx`
nginx errors	`sudo tail -50 /var/log/nginx/error.log`

Troubleshooting

Problem	Check
502 Bad Gateway	`pm2 status` (is hrms-backend online?), `pm2 logs`, and PORT=4000 in .env
App crashes on start, database error	MONGO_URI wrong or this server's IP not allowed in the database firewall
CORS error in the browser	Check the nginx file was saved exactly as in Step 11 and run `sudo nginx -t`
Certbot fails	DNS not pointing here yet, or port 80 blocked
Face login fails	FACE_SERVICE_API_KEY differs between .env and ecosystem.config.js; `pm2 logs hrms-face-service`
scp / rsync says "No such file or directory"	The path was cut by line wrap, or `~` was used for the old server. Use the full `/home/laabhbothra/...` path
pip says "No CMAKE_CXX_COMPILER could be found"	`sudo apt -y install g++ cmake`, then follow Step 8 (dlib-bin first)
"[FATAL] ALLOWED_ORIGINS must be set"	Add it to backend/.env (Step 6), then `pm2 restart hrms-backend --update-env`
Biometric device does not sync	Device server address and port, and the `/iclock/` block in nginx